

Packet Tracer – RIP Routing Basics

Objectives

Part 1: Build a Multi-Router Network Topology

Part 2: Configure IP Addressing

Part 3: Configure RIP Version 2

Part 4: Verify RIP Routing Tables

Part 5: Test Network Reachability

Background / Scenario

As networks grow larger, manually configuring static routes on every router becomes difficult and time-consuming. Dynamic routing protocols solve this problem by allowing routers to automatically learn and share network information.

In this lab, you will configure **Routing Information Protocol (RIP) Version 2** on multiple routers. The routers will exchange routing information automatically, allowing devices on different networks to communicate without manually configured static routes.

This lab focuses on:

- Configuring RIP Version 2
- Advertising networks
- Verifying learned routes
- Testing end-to-end connectivity
- Understanding dynamic routing behavior

You are building a foundation for more advanced routing protocols such as OSPF and EIGRP.

Instructions

Part 1: Build the Network Topology

Step 1: Add the Devices

a. Open **Cisco Packet Tracer**.

b. Add the following devices:

- 3 Routers (1941 or 2911)
- 2 Switches (2960)
- 2 PCs

c. Rename the devices:

- Router0 → **R1**
 - Router1 → **R2**
 - Router2 → **R3**
 - Switch0 → **S1**
 - Switch1 → **S2**
 - PC0 → **PC-A**
 - PC1 → **PC-B**
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Step 2: Connect the Devices

Use **Copper Straight-Through** cables.

Device Interface		Device Interface	
PC-A	FastEthernet0	S1	Fa0/1
S1	Gi0/1	R1	Gi0/0
R1	Gi0/1	R2	Gi0/0
R2	Gi0/1	R3	Gi0/0
R3	Gi0/1	S2	Gi0/1
S2	Fa0/1	PC-B	FastEthernet0

Wait until all links become green.

Network Addressing Plan

Device Interface IP Address

PC-A	Fa0	192.168.1.10 /24
R1	Gi0/0	192.168.1.1 /24
R1	Gi0/1	10.0.12.1 /30
R2	Gi0/0	10.0.12.2 /30
R2	Gi0/1	10.0.23.1 /30
R3	Gi0/0	10.0.23.2 /30
R3	Gi0/1	192.168.3.1 /24
PC-B	Fa0	192.168.3.10 /24

Part 2: Configure IP Addressing

Step 1: Configure PC-A

Go to:

Desktop → IP Configuration

Enter:

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

Step 2: Configure PC-B

Enter:

- IP Address: 192.168.3.10
- Subnet Mask: 255.255.255.0

- Default Gateway: 192.168.3.1

Step 3: Configure R1 Interfaces

enable

configure terminal

interface gigabitethernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface gigabitethernet 0/1

ip address 10.0.12.1 255.255.255.252

no shutdown

exit

Step 4: Configure R2 Interfaces

enable

configure terminal

interface gigabitethernet 0/0

ip address 10.0.12.2 255.255.255.252

no shutdown

exit

interface gigabitethernet 0/1

ip address 10.0.23.1 255.255.255.252

no shutdown

exit

Step 5: Configure R3 Interfaces

enable

configure terminal

```
interface gigabitethernet 0/0
ip address 10.0.23.2 255.255.255.252
no shutdown
exit
```

```
interface gigabitethernet 0/1
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
```

Part 3: Configure RIP Version 2

What Is RIP?

RIP is a dynamic routing protocol that shares network information between routers automatically.

Instead of manually adding routes, routers exchange updates every 30 seconds.

RIP uses:

- Hop Count as its metric
 - Maximum of 15 hops
 - UDP Port 520
 - Automatic route advertisement
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Step 1: Configure RIP on R1

```
router rip
version 2
no auto-summary
network 192.168.1.0
network 10.0.12.0
exit
```

Step 2: Configure RIP on R2

```
router rip
version 2
no auto-summary
network 10.0.12.0
network 10.0.23.0
exit
```

Step 3: Configure RIP on R3

```
router rip
version 2
no auto-summary
network 10.0.23.0
network 192.168.3.0
exit
```

Part 4: Verify RIP Operation

Step 1: Check RIP Configuration

On any router:

```
show ip protocols
```

Expected output:

Routing Protocol is "rip"

Sending updates every 30 seconds

Version control: send version 2

This confirms RIP Version 2 is running.

Step 2: View the Routing Table

On R1:

```
show ip route
```

Expected output:

R 10.0.23.0/30
R 192.168.3.0/24

Notice the routes begin with:

R

The letter **R** indicates the route was learned through RIP.

Step 3: Verify R3 Learned Remote Networks

On R3:

show ip route

Expected output:

R 192.168.1.0/24
R 10.0.12.0/30

R3 has automatically learned the networks behind R1.

Part 5: Test Network Reachability

Step 1: Test Router Connectivity

From R1:

ping 10.0.23.2

The ping should succeed.

Step 2: Ping Between PCs

On PC-A:

ping 192.168.3.10

Expected Result:

Reply from 192.168.3.10

You should receive four successful replies.

Step 3: Ping Back from PC-B

On PC-B:

```
ping 192.168.1.10
```

The ping should also succeed.

Step 4: Trace the Route

On PC-A:

```
tracert 192.168.3.10
```

Expected path:

192.168.1.1

10.0.23.2

192.168.3.10

The packet travels through multiple routers using routes learned from RIP.

Understanding RIP Route Learning

Before RIP:

R1 only knows:

- 192.168.1.0

- 10.0.12.0

After RIP:

R1 learns:

- 10.0.23.0

- 192.168.3.0

Similarly, R3 learns about networks behind R1.

Network Topology

PC-A

192.168.1.10

|
|
S1
|
|
R1
10.0.12.1
|
|
R2
10.0.23.1
|
|
R3
192.168.3.1
|
|
S2
|
|
PC-B
192.168.3.10

RIP automatically exchanges routing information between all routers.

Troubleshooting Tips

If connectivity fails:

✓ Verify interfaces are up:

show ip interface brief

✓ Verify RIP is running:

show ip protocols

✓ Verify routes are learned:

show ip route

✓ Confirm all routers use:

version 2

no auto-summary

✓ Check IP addressing on all interfaces.

✓ Verify the default gateways on both PCs.

✓ Wait approximately 30 seconds after enabling RIP for route advertisements to propagate.

What You Learned

- How dynamic routing differs from static routing.
 - How to configure RIP Version 2.
 - How routers automatically exchange routing information.
 - How to verify RIP using show ip protocols.
 - How to verify learned routes using show ip route.
 - How to test connectivity across multiple routers.
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Challenge Extensions (Optional)

Try modifying the lab:

1. Add a fourth router and another network.
2. Configure RIP on the new router and observe route propagation.
3. Use Simulation Mode to watch RIP update packets.
4. Shut down one router interface and observe how routing changes.
5. Compare RIP with Static Routing from the previous lab.
6. Replace RIP with OSPF and compare the routing tables.